Power system:

Power Distribution Architecture:

Our Power distribution is relatively simple. The main voltages used will be 8.4-7.4v provided by our 2s LiPo battery, and 5v provided by our buck converter and our SBC/SBM.
7v4-8v4 → Motor driver, Buck converter.
5V → Raspberry pi & Nano, Sensors, Motor driver power, Servo motor.
3v3 → Cam (supplied by Pi/ribbon cable), May components (i.e All ToF, I2C multiplexer) are compatible with both 3v3 and 5V.
Common ground will be the negative terminal of our LiPo pack to provide a common fixed voltage ref.

Power consumption analysis:

Assuming an unideal scenario:

Hardware	Voltage	Current draw
Raspberry pi zero W at 100% cpu capacity	5v	~220mA
Camera	5v	200-225mA
N20 motor	7.4v motor, 3v3 or 5v logic	180-550mAh usage
Arduino Nano (UART connection to PI)	3v3	30-50mA
All 6 VL53L1X (TBD)	5V-3v3	120mA (20mA per)
2 VL53L3CX (TBD)	3v3-5V	40mA (20mA per)
TB6612FNG driver	7.4v for motors, 5v consumption	5mA
TCA9548A	3v3-5V	~1mA, negligible
TOTAL	N/A	796mA - 1.2A draw

With a 350mAh battery, battery life can be expected to be:
No current spikes → 26min 15 seconds battery life.
Infrequent current spikes → ~22-24 minute battery life.
By the term current spikes, we refer mainly to when the n20 motor is suddenly loaded. Given the environment, current spikes will most likely only appear when the bot starts moving from a standstill and when our code malfunctions and knocks into an obstacle. Current spikes will be rare, but it doesn't hurt to account for them.